THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If x° and 24° are complementary, then $x =$ 1

(a) 86° **(b)** 56° **(c)** 66° **(d)** 76°

2. A two-digit number has digits 2 and 5. The sum of the number and its reverse is: 1

(a) 76 **(b)** 77 **(c)** 78 **(d)** 79

3. Two sides of a triangle are 5 cm and 9 cm. The third side could be: 1

(a) 16 cm **(b)** 14 cm **(c)** 7 cm **(d)** 4 cm

4. How many pieces of length $\frac{1}{4}$ m can be cut from a rope of 3 m? 1

(a) 6 **(b)** 24 **(c)** $\frac{3}{4}$ **(d)** 12

5. In a linear pair one angle is 2 times the other. The smaller angle is: 1

(a) 80° **(b)** 50° **(c)** 70° **(d)** 60°

6. A two-digit number has digits 9 (tens) and 2 (units). Number minus its reverse is: 1

(a) 65 **(b)** 64 **(c)** 62 **(d)** 63

7. The angles of a triangle are x°, $2x^\circ$ and $3x^\circ$. The value of x is: 1

(a) 29 **(b)** 31 **(c)** 30 **(d)** 32

8. Scaling a drawing by $\frac{3}{4}$: a line of 20 cm becomes (cm): 1

(a) $7\frac{1}{2}$ **(b)** 15 **(c)** $26\frac{2}{3}$ **(d)** 30

9. Two parallel lines are cut by a transversal. An interior angle is 65°. The interior angle on the same side is: 1

(a) 105° **(b)** 125° **(c)** 135° **(d)** 115°

10. The number of steps the Collatz sequence from 5 takes to reach 1 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

11. The exterior angle of a triangle is $(3x + 10)^\circ$ and the interior opposite angles are 50° and 50°. The value of x is: 1

(a) 31 **(b)** 29 **(c)** 32 **(d)** 30

12. The number of $\frac{1}{4}$s in 2 is: 1

(a) 8 **(b)** 2 **(c)** 4 **(d)** 6

13. If a transversal makes equal corresponding angles with two lines, the lines are: 1

(a) equal **(b)** intersecting **(c)** parallel **(d)** perpendicular

14. The smallest digit $*$ making $4*3$ divisible by 9 is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

15. If $\triangle ABC \cong \triangle PQR$, then side BC equals: 1

(a) AB **(b)** PR **(c)** QR **(d)** PQ

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Two supplementary angles are such that one is 2 times the other. Find them. 2

17. Show that $(10a + b) + (10b + a) = 11(a + b)$ and verify for $a = 3$, $b = 5$. 2

18. Two sides of a triangle are 6 cm and 10 cm. Find the least and the greatest whole-number length of the third side. 2

19. How many $\frac{1}{8}$s are there in $\frac{3}{4}$? 2

20. A transversal cuts two lines and makes corresponding angles of 75° and 75°. Are the lines parallel? Give a reason. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Three of the four angles around a point are 95°, 85° and 110°. Find the fourth angle and name the kind of each of the four angles. 3

22. Apply Kaprekar's routine twice to 5296. 3

23. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 10)^\circ$ and $\angle B = (2x - 5)^\circ$. Find x and all three angles. 3

24. A rectangle has sides $\frac{3}{4}$ m and $\frac{2}{5}$ m. Show its area as the product of the sides and compare it with the area of the smaller side squared. 3

OR

A transversal cuts two lines and makes corresponding angles of 65° and 65°. Are the lines parallel? Give a reason. 3

25. Find how many reverse-and-add steps turn 57 into a palindrome, showing each step. 3

OR

In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 5$ cm, $BC = QR = 7$ cm and $\angle B = \angle Q = 50^\circ$. Name the congruence rule, and find PR if $AC = 9$ cm. Find also the perimeter of $\triangle PQR$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A transversal cuts two parallel lines and one of the eight angles is 50°. How many of the eight angles measure 50° and how many measure 130°? Give reasons. 5

OR

For the three-digit number 732, find its reverse, the difference, and show that the difference is divisible by 99. 5

27. The angles of a triangle are $(x + 10)^\circ$, $(x - 10)^\circ$ and $(2x + 20)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). **5**

OR

A journey of 120 km is done in three parts. The first part is $\frac{1}{4}$ of the journey; $\frac{2}{3}$ of the remaining distance is covered next. Find both distances and the distance still left. **5**

END OF QUESTION PAPER **Total: 50 Marks**

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